

艾滋病患者隐球菌感染筛查 浙江省专家共识

浙江省医学会热带病和寄生虫病分会艾滋病学组

通信作者:朱彪,浙江大学医学院附属第一医院感染科,传染病诊治国家重点实验室,感染性疾病诊治协同创新中心,310003 杭州,Email:zhubiao1207@zju.edu.cn

【摘要】 隐球菌感染是艾滋病相关死亡最常见的原因之一。早期筛查发现隐球菌感染,提前进行抗隐球菌治疗,可明显减少艾滋病患者的病死率。我们组织相关专家制订了本共识。本共识推荐:(1)CD4⁺T 淋巴细胞数 <200 个/ μ L 的艾滋病患者应进行隐球菌抗原筛查;(2)侧向免疫层析法为艾滋病患者隐球菌感染筛查的首选方法;(3)血隐球菌抗原筛查阳性的艾滋病患者需进行脑脊液隐球菌抗原检测;(4)有条件的地区可开展血隐球菌抗原定量检测;(5)无症状的隐球菌抗原血症的艾滋病患者建议进行抢先的氟康唑抗隐球菌治疗;(6)血隐球菌抗原阳性,同时脑脊液隐球菌抗原阳性、隐球菌败血症、肺隐球菌病和其他部位隐球菌病的艾滋病患者,需按相关指南进行规范抗隐球菌治疗。

【关键词】 隐球菌病; 获得性免疫缺陷综合征; 筛查; 治疗; 共识

基金项目:“十三五”国家科技重大专项(2017ZX10202102,2018ZX10715014)

DOI:10.3760/cma.j.issn.1674-2397.2019.02.001

Expert consensus on screening for cryptococcal infection among AIDS patients in Zhejiang

AIDS Group, Society of Tropical Diseases and Parasitology of Zhejiang Medical Association

Corresponding author: Zhu Biao, Department of Infectious Diseases, State Key Laboratory for Diagnosis and Treatment of Infectious Diseases, Collaborative Innovation Center for Diagnosis and Treatment of Infectious Diseases, the First Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou 310003, Email: zhubiao1207@zju.edu.cn

【Abstract】 Cryptococcal infection is one of the most common causes of AIDS-related death. Early screening and early anti-cryptococcal treatment can significantly reduce the mortality of AIDS patients. This expert consensus recommends: 1. Patients with CD4⁺T lymphocyte <200/ μ L should be screened for cryptococcal antigen; 2. Lateral-flow assay should be the first choice for screening cryptococcus infection in AIDS patients; 3. HIV patients with positive serum cryptococcal antigen (CrAg) in screening should be tested for CrAg in cerebrospinal fluid (CSF); 4. The quantitative detection of CrAg can be carried out if feasible; 5. AIDS patients with asymptomatic cryptococcal antigenemia should receive preemptive fluconazole anticyptococcal therapy; 6. AIDS patients with positive serum CrAg, and positive CSF CrAg, cryptococcal septicemia, pulmonary cryptococcal disease or cryptococcal disease in other parts should receive standardized therapy according to the relevant anti-cryptococcal guidelines.

【Key words】 Cryptococcosis; Acquired immunodeficiency syndrome; Screening; Treatment; Consensus

Fund program: National Key Technologies R&D Program of China during the 13th Five-Year Plan Period (2017ZX10202102, 2018ZX10715014)

DOI:10.3760/cma.j.issn.1674-2397.2019.02.001

隐球菌感染是艾滋病患者最常见的机会性感染

引用格式:浙江省医学会热带病和寄生虫病分会艾滋病学组. 艾滋病患者隐球菌感染筛查浙江省专家共识[J]. 中华临床感染病杂志, 2019, 12(2):81-86, 106. DOI:10.3760/cma.j.issn.1674-2397. 2019.02.001.

之一。艾滋病患者合并隐球菌感染时,与普通人群相比,无论在发病率、临床表现还是预后,都存在较大差异。近年来,隐球菌感染的筛查方法有了较大改进,隐球菌抗原筛查已被写入多部艾滋病诊治指南,多项多中心、大样本的临床研究对此进行了临床

经济学评价^[1-3]。但我国尚缺乏艾滋病患者隐球菌感染筛查规范,因此我们组织有关专家制订了本共识,供各位同道参考。

1 隐球菌感染筛查的意义

近年来,人类免疫缺陷病毒(HIV)感染者逐年增加。根据联合国艾滋病规划署(UNAIDS)统计,至2017年,全球HIV感染者约3 690万(3 011万~4 039万),2017年艾滋病相关死亡人数约为94万(67万~130万)。HIV感染者中,7.1%~19.0%伴有隐球菌感染^[4-7]。我国HIV感染者尚缺乏隐球菌感染完整的流行病学数据,但从1986至2006年发表的文献看,我国隐球菌感染者数量在逐年攀升^[8],据文献报道,我国HIV感染者中隐球菌抗原阳性率为9.8%~62.3%,各地报道差异较大^[9-10]。

隐球菌感染是艾滋病相关死亡中最常见的原因之一。其中因隐球菌脑膜炎导致艾滋病患者相关死亡约占15%~41%^[11-15],常见表现为发热、头痛、嗜睡、人格改变、记忆丧失及颈项强直等亚急性或慢性脑膜脑炎的症状和体征。HIV感染人群并发隐球菌脑膜炎通常脑脊液具有更高的真菌负荷量,更易发生播散性感染,隐球菌培养阳性率更高,临床表现危重且不典型^[16-17]。未经治疗的隐球菌感染患者病死率约为75%,其中CD4⁺T淋巴细胞<100个/ μ L的患者病死率更高,可高达100%^[18]。颅内、肺部、皮肤、骨、眼睛及前列腺等部位都是隐球菌感染的靶器官。

隐球菌感染者,早期无任何临床症状和影像学改变,仅表现为血隐球菌荚膜多糖抗原检测阳性,即无症状隐球菌抗原血症。在隐球菌感染的早期阶段或隐性感染阶段,这些感染者未经干预多数可发展为显性感染,约1/3无症状的隐球菌抗原血症患者可同时出现无症状的隐球菌脑膜炎^[19],而无症状的隐球菌抗原血症发展为有临床表现的隐球菌病的中位数时间约为22 d^[20-21]。隐球菌抗原阳性的HIV感染者1年内病死率约为39.7%,隐球菌抗原阴性的HIV感染者病死率约为13.9%^[22]。此外,艾滋病合并隐球菌感染在启动高效抗反转录病毒治疗(Highly active antiretroviral therapy, HAART)后可出现免疫重建炎症反应综合征^[23],无症状的隐球菌抗原血症患者在未抗隐球菌治疗的情况下启动HAART,最初3个月的病死率增加18%^[4, 20]。因此,尽早筛查出这些无症状隐球菌抗原血症患者,进行抗隐球菌治疗,可大幅减少HIV感染者的病

死率。

隐球菌抗原的筛查还可节约有限的医疗资源,与普遍性的氟康唑预防性治疗相比,具有更好的成本效益。有研究表明,对于CD4⁺T淋巴细胞数<100个/ μ L的HIV感染者进行隐球菌抗原筛查,提前进行抗隐球菌治疗,虽会增加筛查费用,但总体医疗成本降低,感染者的预后得到明显改善^[24-26]。

2 隐球菌感染筛查方法

隐球菌感染检测的实验方法有隐球菌免疫学检测、体液涂片墨汁染色、组织病理学诊断、影像学筛查、隐球菌培养和分子生物学检测等。

2.1 隐球菌免疫学检测 隐球菌免疫学检测发展迅速,主要有3种方法:侧向免疫层析法(Lateral-flow assay, LFA)、乳胶凝集试验(Latex agglutination assay, LA)和酶免疫分析法(Enzyme immunoassay, EIA)。LFA方法简单快速,应用最为广泛。

2.1.1 LFA 是以胶体金标记单克隆抗体检测新型隐球菌的全部4种荚膜多糖抗原。对隐球菌脑膜炎患者来说,脑脊液LFA的敏感度为94%~100%。血清LFA与脑脊液LFA在检测敏感度、特异度、阳性预测价值和阴性预测价值等方面都基本一致,在全血、血浆和血清中检测结果均高度一致。尽管LFA诊断隐球菌感染的总体敏感度和特异度分别为97.6%和98.1%,但对于孤立性肺隐球菌病检测的敏感度不足^[27]。LFA的假阴性和假阳性率低,疾病早期,溶血或极高浓度的隐球菌抗原可出现假阴性,出现假阴性结果时应进行样本稀释,用半定量检测法来排除假阴性;LFA与曲霉有抗原交叉反应,可出现假阳性结果。

由于LFA操作简单,检测速度快(<15 min),技术简单,所需实验仪器少,同时试剂不需要冷藏,已替代传统的隐球菌感染筛查方法。

2.1.2 LA 是以乳胶颗粒为载体,表面结合高效价的抗隐球菌荚膜多糖抗体,可吸附样本中的相应抗原,形成肉眼可见的凝集反应颗粒。文献报道,血LA的敏感度为93%~100%,特异度为93%~98.6%,阳性预测价值为92.4%,阴性预测价值为99.2%^[28-30]。临床检测中会出现假阳性和假阴性的情况:某些风湿免疫性疾病,如类风湿因子关节炎、系统性红斑狼疮、结节病及巨球蛋白血症等,可引起假阳性,同时二氧化碳噬纤维菌属、毛孢子菌属、曲霉和抗酸杆菌的抗原均可与新型隐球菌的荚膜多糖抗原存在交叉反应,可导致假阳性反应。隐

球菌滴度过高或过低,感染早期阶段和不合适的标本存储可引起假阴性反应^[31-32]。

2.1.3 EIA 通过检测抗隐球菌抗体来确认是否存在隐球菌感染,其特异度及敏感度与 LA 相似。但 EIA 步骤繁多,对实验室要求高,耗时长,易产生误差。并且对既往感染者与定植菌感染者无法鉴别,易造成假阳性,当患者处于疾病早期或免疫功能极度低下时,抗体产生不足,则容易造成假阴性。

2.2 脑脊液墨汁染色涂片 隐球菌检查的常用方法是墨汁染色找隐球菌,其敏感度取决于隐球菌的负荷量,HIV 患者中敏感度可达 80% 左右^[33]。

2.3 组织病理学检查 组织病理学可通过对肺、皮肤、骨髓、脑等多种组织进行检测,比墨汁染色敏感度更高^[34]。急性期组织学改变常以凝固性坏死、形成脓肿与炎性细胞浸润为主,慢性期一般表现为肉芽肿性炎或伴坏死和纤维结缔组织增生,HE 染色下,隐球菌菌体圆形,呈淡蓝或灰红色,菌体周围空隙为荚膜收缩所致;淀粉酶消化后碘酸雪夫、黏液卡红、六胺银染色等特殊染色,可使隐球菌孢子分别显色为紫红色、鲜红色和棕褐色^[35]。

2.4 隐球菌感染影像学检查 艾滋病患者并发肺隐球菌肺炎,患者肺部影像学检查常表现为结节团块,缺乏特征性表现,易被误诊为肿瘤、细菌性肺炎、结核、真菌性肺炎。弥漫播散性肺炎表现为浸润、实变和空洞形成、晕征、弥漫粟粒影及磨玻璃样渗出等^[27]。肺部增强 CT 扫描示,病灶内部多呈不均匀强化。

2.5 隐球菌培养 隐球菌培养是隐球菌感染的确诊手段。对艾滋病合并隐球菌脑膜炎患者,脑脊液和血液隐球菌培养阳性率分别为 90% 和 70%^[36]。培养阳性者可结合药敏试验,指导治疗。

2.6 分子生物学检测 隐球菌分子生物学检测方法有多种,包括聚合酶链反应 (Polymerase chain reaction, PCR)、环介导等温扩增技术等。分子生物学检查可快速特异地检出隐球菌感染,可用于隐球菌的菌种鉴定、分型,但对检测设备及技术人员要求较高,临床目前未常规开展。

推荐意见一: LFA 为艾滋病隐球菌感染筛查的首选方法。

3 隐球菌感染筛查策略

3.1 隐球菌感染筛查人群 国外研究发现,CD4⁺T 淋巴细胞 < 100 个/ μ L 的 HIV 感染者中隐球菌抗原阳性率为 6.5% ~ 21.0%, CD4⁺T 淋巴细胞为 101 ~

200 个/ μ L 的 HIV 感染者中隐球菌抗原阳性率为 2% ~ 6%。18.6% 的隐球菌抗原阳性者 CD4⁺T 淋巴细胞为 100 ~ 200 个/ μ L (95% CI = 15.4% ~ 22.2%)。CD4⁺T 淋巴细胞 < 100 个/ μ L 的患者中,住院患者隐球菌抗原阳性率为 9.8%,门诊患者为 6.3%^[4, 22, 37]。对 CD4⁺T 淋巴细胞 < 100 个/ μ L 的艾滋病患者在启动 HAART 或重启 HAART 前,进行隐球菌抗原筛查,有更多的临床获益和更好的成本效益^[38-39]。世界卫生组织 (WHO) 指南中,强烈建议对于 CD4⁺T 淋巴细胞 < 100 个/ μ L 的 HIV 感染者在未启动 HAART 前进行隐球菌抗原筛查^[1]。

有研究发现,在 CD4⁺T 淋巴细胞 < 100 个/ μ L 和 101 ~ 200 个/ μ L 的人群之间,隐球菌抗原阳性率无统计学意义^[4]。国外一项纳入 5 461 例患者的回顾性研究发现,在 CD4⁺T 淋巴细胞 < 100 个/ μ L 的隐球菌抗原阳性患者中出现隐球菌相关疾病的阳性预测价值为 96%,而 CD4⁺T 淋巴细胞为 101 ~ 200 个/ μ L 的患者中阳性预测价值则高达 100%^[40]。因此,如仅对 CD4⁺T 淋巴细胞 < 100 个/ μ L 的患者进行隐球菌抗原筛查存在较高的漏检率^[41],对于 CD4⁺T 淋巴细胞为 100 ~ 200 个/ μ L 的患者进行常规的隐球菌抗原筛查,可有潜在受益。

推荐意见二: CD4⁺T 淋巴细胞 < 200 个/ μ L 的艾滋病患者应进行隐球菌抗原筛查^[1, 4, 30, 37, 42-44]。

3.2 隐球菌抗原血症患者的进一步检查 血隐球菌抗原阳性者需根据临床表现进一步检查,及时发现隐球菌感染累及的部位,包括腰穿脑脊液检查,肺部 CT 及其他部位的影像学检查等,应常规进行血和脑脊液实验室快速筛查,包括隐球菌抗原检测、墨汁染色及隐球菌培养。在实验室条件受限地区,可采用实时 LFA 进行隐球菌的床边筛查^[1]。如条件允许,可对可疑的感染病灶进行组织学活检或培养,以进一步明确有无隐球菌感染。

推荐意见三: 血隐球菌抗原筛查阳性的艾滋病患者需进行脑脊液隐球菌抗原检测。

3.3 血隐球菌抗原定量检测 有研究发现,隐球菌抗原滴度和患者的病死率及隐球菌脑膜炎发生率密切相关^[45-46]。研究认为隐球菌抗原滴度 > 1:512 和播散性隐球菌病密切相关^[40]。2010 年 WHO 指南提出隐球菌血症的患者,当隐球菌抗原滴度 > 1:512 要按隐球菌脑膜炎治疗^[47]。但国内外多项研究结论差异较大^[41, 48-49]。2017 年,一项大规模多中心的临床研究显示,19 233 例 CD4⁺T 淋巴细胞 < 100 个/ μ L 的 HIV 感染者进行血隐球菌抗原筛查,

发现血隐球菌抗原滴度 $>1:160$ 可预测并发无症状的隐球菌脑膜炎(敏感度为 88.2%, 特异度为 82.1%)^[50]。在无法进行腰穿检查的地区,血隐球菌抗原滴度可用来推测易发生隐球菌脑膜炎的高危人群。其价值尚需大规模的临床研究来进一步验证。

推荐意见四:有条件的地区可开展血隐球菌抗原定量检测。

4 隐球菌抗原筛查后的治疗策略

4.1 血隐球菌抗原阳性,脑脊液隐球菌抗原阳性患者 按照隐球菌脑膜炎进行治疗,先进行两性霉素 B($0.5 \sim 0.7 \text{ mg} \cdot \text{kg}^{-1} \cdot \text{d}^{-1}$) + 氟胞嘧啶($100 \text{ mg} \cdot \text{kg}^{-1} \cdot \text{d}^{-1}$)的 4 周诱导治疗,后进行氟康唑 $600 \sim 800 \text{ mg/d}$ 和(或)氟胞嘧啶 $100 \text{ mg} \cdot \text{kg}^{-1} \cdot \text{d}^{-1}$ (≥ 6 周)的巩固治疗和氟康唑 200 mg/d (≥ 1 年)的维持治疗^[51],抗真菌 4 周后启动 HAART;其他替代方案参考相关国家指南。

4.2 血隐球菌抗原阳性,脑脊液隐球菌抗原阴性患者 进行肺部 CT 筛查,临床和影像学怀疑肺隐球菌病者,应进行抗隐球菌治疗,短期复查,有效继续治疗,无效进行肺组织活检。治疗方案:(1)无症状、轻中度症状、无肺部播散性感染者,给予氟康唑 400 mg/d , 6 ~ 12 个月;(2)重度症状患者,肺部播散性感染按隐球菌脑膜炎治疗^[47],抗真菌治疗 2 周后启动 HAART^[1]。

4.3 血隐球菌抗原阳性的患者 若脑脊液隐球菌抗原阴性,无脑膜炎表现,肺部无肺隐球菌病依据者:(1)如有隐球菌败血症,应按照隐球菌脑膜炎治疗;(2)无隐球菌败血症或单个部位感染者,应使用氟康唑 400 mg/d , 6 ~ 12 个月^[47],在抗真菌治疗 2 周后启动 HAART。

4.4 无症状隐球菌抗原血症患者 要进行抢先的抗真菌治疗,先进行 2 周的氟康唑 800 mg/d 诱导治疗,再进行 8 周的氟康唑 400 mg/d 巩固治疗,在启动 HAART 期间巩固治疗后进行氟康唑 200 mg/d 维持治疗^[1],抗真菌治疗 2 周后启动 HAART。

4.5 由于血液、脑脊液中隐球菌抗原会持续阳性数月甚至数年,因此对于有隐球菌脑膜炎病史的患者不需要进行常规隐球菌抗原筛查^[52]。若既往有隐球菌脑膜炎病史的患者,出现了新的脑膜炎症状,需再次进行腰穿,进行脑脊液隐球菌培养,并进行药敏试验,全面评估是否出现隐球菌的复发^[1]。

4.6 隐球菌抗原阳性的育龄期妇女 在氟康唑

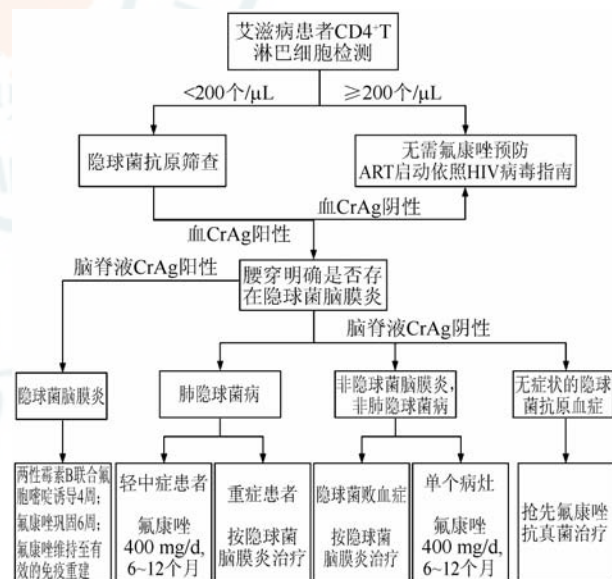
(妊娠 D 级)治疗前应进行妊娠试验。对于隐球菌抗原阳性的孕妇需进行腰穿,如脑脊液隐球菌抗原阳性,需进行两性霉素 B 治疗。如脑脊液隐球菌抗原阴性,则权衡氟康唑抗隐球菌治疗的利弊,由经验丰富的艾滋病专家制订方案^[1]。

4.7 抗隐球菌治疗疗程推荐 (1)可进行 HIV 病毒载量检测:在 HAART 治疗稳定,抗隐球菌维持治疗 >1 年, CD4^+ T 淋巴细胞 >100 个/ μL ,病毒载量检测不到,可考虑停药;(2)若无法进行 HIV 病毒载量检测,在 HAART 治疗稳定,抗隐球菌维持治疗 >1 年, CD4^+ T 淋巴细胞 >200 个/ μL ,可考虑停药^[1]。

推荐意见五:无症状的隐球菌抗原血症的艾滋病患者建议进行抢先的氟康唑抗隐球菌治疗。

推荐意见六:血隐球菌抗原阳性,同时脑脊液隐球菌抗原阳性、隐球菌败血症、肺隐球菌病和其他部位隐球菌病的艾滋病患者,按相关指南进行规范抗隐球菌治疗。

5 艾滋病患者隐球菌抗原筛查流程图



注:CrAg. 隐球菌抗原;ART. 抗反转录病毒治疗

利益冲突 所有作者均声明不存在利益冲突

参与共识编写专家(按姓氏汉语拼音排序):

鲍志坚 房英杰 高中华 葛宇黎 郭永征 洪亮 胡耀仁
黄建荣 黄冠成 金灵肖 居红珍 李世波 李晓峰 李月翠
凌乔 楼莲青 卢伟力 罗艳 马瞧勤 邱济海 邱志红
沈巧凤 沈显元 石勇明 苏菲菲 陶然 童凤军 童永喜
王大勇 王炜 王章善 吴南屏 夏小学 肖强古 许烂漫
许利军 宣华 颜春晖 杨春 杨杰 杨劲 叶荣夏
喻剑华 张德和 张海云 郑南红 周春华 朱彪 朱梦飞

学术秘书兼执笔:陶然 郭永征

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- (收稿日期:2019-03-05)
(本文编辑:金建华)
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- (收稿日期:2018-12-27)
(本文编辑:金建华)